

steps towards a routing enabled road network

I. Creating the “Transport Graph”

1. import all highways with type, name and surface tags as attributes
2. add the `area_type` attribute for all non-road-types by
 1. find the `st_intersects` spatial relation to enclosing features and
 1. add the “landuse” tag
 2. add the “leisure” tag
 3. add the tags as arrays to compensate for paths crossing several type areas
3. Prune the road network based on
 1. road type
 2. area type
 3. e.g.: footpath in non-residential areas like cemeteries / parks, cycle way
 4. ruleset: tbd
 5. doubling-tag:
 1. one-sided = 0
 2. to double = 1
4. Duplication
 1. duplicate all edges with tag = 1
 2. construct intersections with all edges having tag = 0

Functions and order of execution

1. `create_edges_and_vertices => 00_create_edges_table.sql`
2. `chop_lines_and_extract_nodes => chop_lines_and_extract_source_nodes_function_test.sql`

II. Creating the “Access Graph”

1. build the junctions from road to address
 1. find the nearest edge
 2. graft the touchpoint into the edge
 3. segment the enhanced edge

III. Create a simplified logical graph

1. identify nodes having degree = 2
2. collapse contiguous edges consisting of such nodes into one logical edge
 1. add the length of the constituent edges
 2. collect the `node_ids` with degree 2 into an array ordered by path index
3. collect the edge geometries into a geometry collection of linestring primitives

The result of this sketched out process will be a detailed road network graph. In residential areas it will be a fine grained street side differentiating routing network. Those fine grained near-terminus parts will be connected with far ranging logical edges in non-residential areas.

The routing will be calculated on the logical edges only. The dataset will preserve the prior geometry as a geometry array or multigeometry, thus combining a logical network with its physical representation within one data structure.

Steps towards the universal graph

What kind of edges do we create in the process?

source	target
road segment	duplicate edges left and right
vertex with degree 1	a simple bridge
vertex with degree 2	a simple bridge
vertex with degree 3 or more	a number of bridges equal to the vertex' degree
vertex from "closest point to PoI"	a simple edge

1. create point junctions

1. Input:

- points of interest
- road network

2. Functions:

- asd
- asd

3. Result

- nodes in provisory node table with type "junction"
 - node_id
 - geom
 - type = junction
 - properties
 - * way_id
 - * poi_type

2. graft nodes into road segments

3. create all new nodes

1. Input:

- points of interest
- road network

2. Functions:

3. Result

4. create all new edges

1. Input:

- points of interest
- road network

2. Functions:

3. Result

```
graph TD;
  A-->B;
  A-->C;
```

```

    B-->D;
    C-->D;

graph TD;

A(addresses)
B(Access Points)
C(other PoI)

D{create<br>junctions}

E(road_network)

F(enhanced road<br>linestring)

G(Segments)

H(nodes)

A-->D
B-->D
C-->D

D--graft junction into road segment-->F

E-->F

F--Split linestring<br>into segments-->G

G--create child nodes<br>from each edge-->H

```